ESPaperPlay / DEVELOPMENT BOARD r1

ESP32-S3-WROOM-1-N16R8 / GDEY075T7-T01 / protected 1-cell battery

Six editable functional sheets. DESIGN REVIEW ONLY; no PCB release.

POWER_INPUT_CHARGER

Input / charger / battery

文件: POWER_INPUT_CHARGER.kicad_sch

POWER_3V3_CONTROL

3.3 V / IO expander / gauge

文件: POWER_3V3_CONTROL.kicad_sch

MCU_CORE

N16R8 module / reset / buttons

文件: MCU_CORE.kicad_sch

USB_INPUT

USB-C / protection / source select

文件: USB_INPUT.kicad_sch

POWER_DOMAINS

Five switched rails / current links

文件: POWER_DOMAINS.kicad_sch

TOUCH_ISOLATION

GT911 four-signal isolation

文件: TOUCH_ISOLATION.kicad_sch

ASSEMBLY / REVIEW BOUNDARIES

R108 CHG_ARM: DNP. Actual battery / NTC / charge window must be approved before charging.
J502 source selector: one shunt only; J202 MAIN_RUN: one shunt only.
EPD boost + FPC, SD socket and audio ICs remain separate next-stage work.
Power-part ratings / DLA footprint / connector pinouts remain open; no measured performance.

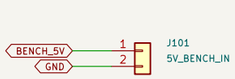
Read hardware/devboard-r0/README.md and docs/v2/hardware/power-design-spec.md before assembly.

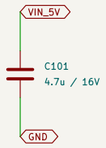
DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL
ESPaperPlay / xiaobaoaaaaaa

Sheet: /		
File: ESPaperPlay_Devboard.kicad_sch		
Title: ESPaperPlay DEVELOPMENT BOARD — PLATFORM DRAFT		
Size: A3	Date: 2026-11-02	**Rev: r1 DRAFT**
KiCad E.D.A. 10.0.7		Id: 1/7

1 POWER PATH AND PROTECTED BATTERY INTERFACE

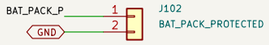
5 V only. USB_INPUT selects protected USB or J101 BENCH_5V into VIN_5V.

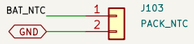




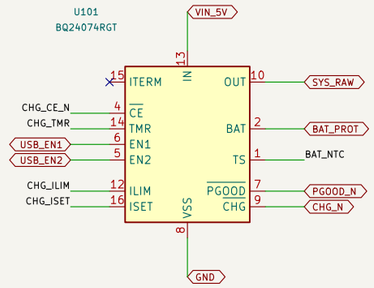
Known, current-limited 5 V bench source.
Select at J502; never bridge both sources.

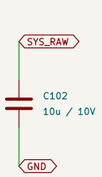


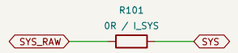


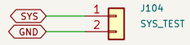


BAT + NTC are functional bench terminals.
Pack connector / polarity / R-T curve TBD.
No fixed resistor in place of pack NTC.





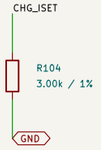




R101: measure total SYS load.
CE pull-up stays on SYS_RAW side.









ISET / ILIM / TMR = calculation starting points, not approved charging settings.
ITERM pin 15 NC: internal termination threshold (mode dependent).

3 AON-SAFE VBUS SENSE



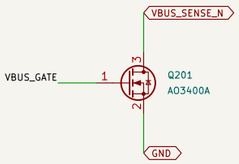


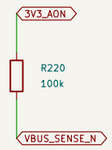




PWR_FLAG: normal RUN with R101/R203 populated.
ERC does not model removable links or input protection.



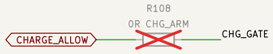


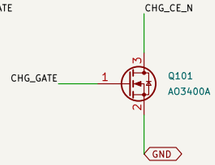




LOW = protected USB branch powered.
PGOOD reports selected input; PTC trip differs.

2 HARDWARE CHARGE INHIBIT / CURRENT MODE / STATUS







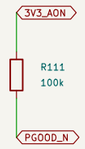








EN2 EN1: 00=100mA, 01=500mA,
10=RILIM, 11=USB suspend.
Do not infer source capacity from VBUS.





J105 normally OPEN. Recovery only: MAIN_RUN OFF, AMP disabled,
EN1/EN2=00, qualified pack + NTC, known bench source.
R108 is DNP: firmware cannot enable charging in initial assembly.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

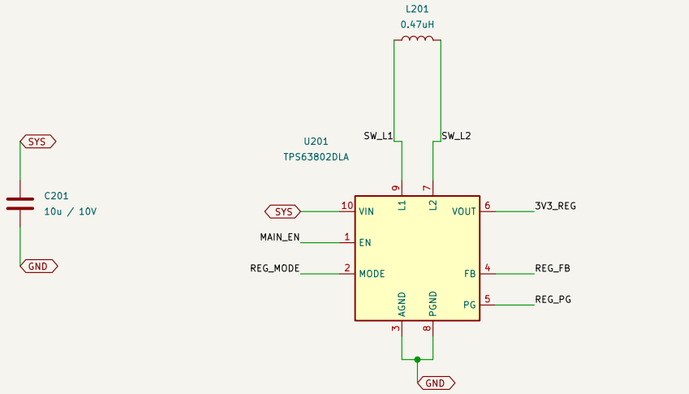
ESPaperPlay / xiaobaoaaaaaa

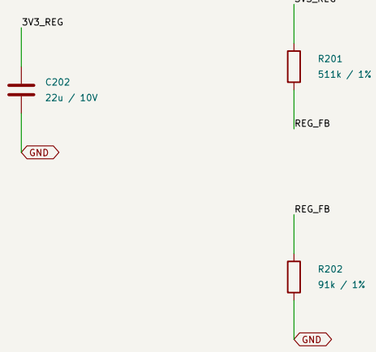
Sheet: /POWER_INPUT_CHARGER/
File: POWER_INPUT_CHARGER.kicad_sch

Title: POWER INPUT / BQ24074 CHARGER

Size: A3	Date: 2026-11-02	**Rev: r1 DRAFT**
KiCad E.D.A. 10.0.7		Id: 2/7

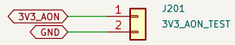
1 MAIN BUCK-BOOST / MEASUREMENT / RECOVERY

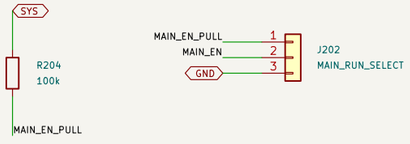


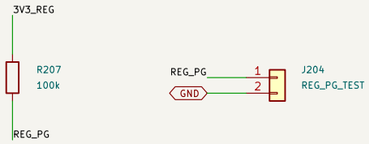




Never parallel an external 3.3 V source.
Open R203 before external injection.
C202 + R201 remain upstream of R203.



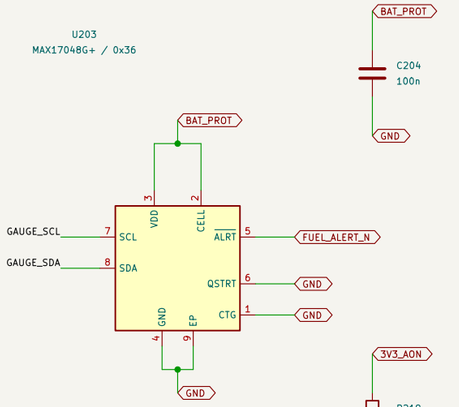




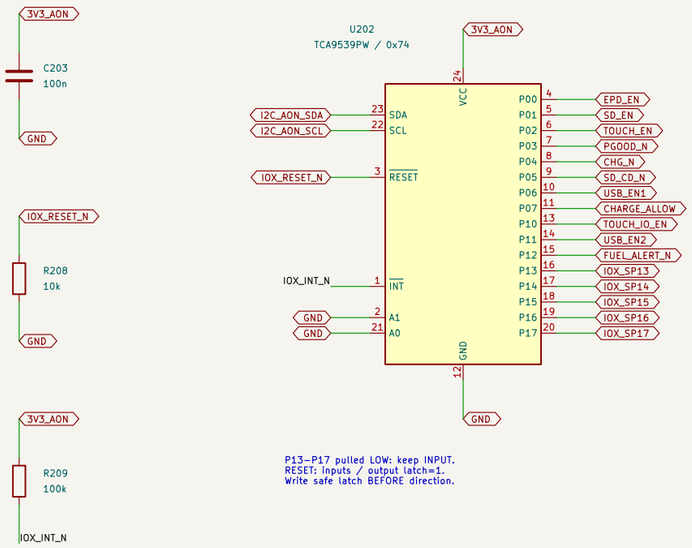
J202: 1-2 RUN; 2-3 OFF / recovery.
One shunt only. Open defaults OFF.

MODE low = automatic PFM/PWM.
U201 exact DLA footprint not assigned.
PG is a test signal, not a new MCU GPIO.

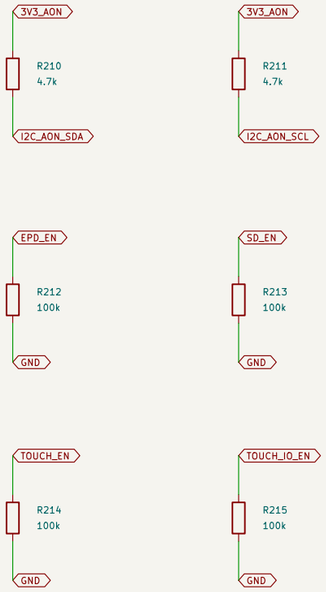
3 BATTERY GAUGE

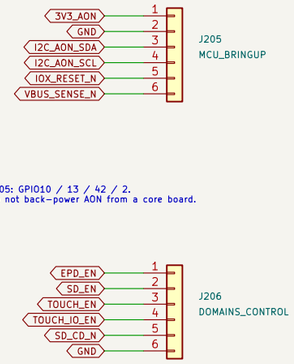


2 AON I2C CONTROL / RESET DEFAULTS



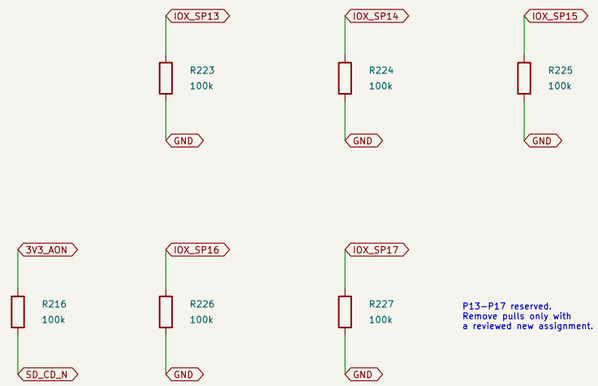
P13-P17 pulled LOW: keep INPUT.
RESET: inputs / output latch=1.
Write safe latch BEFORE direction.





J205: GPIO10 / 13 / 42 / 2.
Do not back-power AON from a core board.

4 SPARE PORT DEFAULTS



P13-P17 reserved.
Remove pulls only with
a reviewed new assignment.

MCU, load switches and touch isolation are now connected across sheets.
Start I2C at 100kHz; edge timing / gauge power-off leakage to verify.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

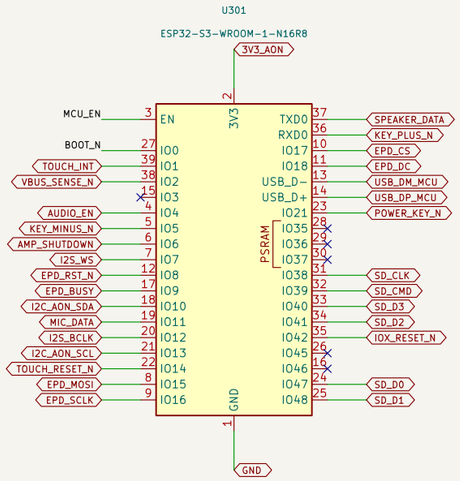
ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_3V3_CONTROL/
File: POWER_3V3_CONTROL.kicad_sch

Title: 3.3 V REGULATOR / POWER CONTROL / GAUGE

Size: A3	Date: 2026-11-02	**Rev: r1 DRAFT**
KiCad E.D.A. 10.0.7		Id: 3/7

1 MCU / N16R8 PHYSICAL PAD MAPPING



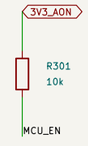
NC: GPIO3 / 45 / 46 (straps); GPIO35 / 36 / 37 (PSRAM).
Do not substitute WROOM-1-N16R16V: pad voltage differs.
Pads 24 / 25 = GPIO47 / 48, respectively.
Antenna at PCB edge; final keepout per module land pattern.

LOCAL BYPASS



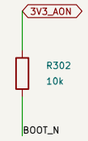


3 MAINTENANCE RESET / DOWNLOAD



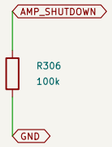




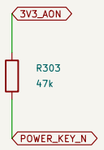




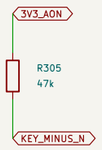
EN: 10k / 1uF initial RC; validate actual power ramp.
BOOT: pull-up only, no large capacitor.
GPIO43 / 44 are assigned: no always-connected UART0.
Native USB Serial/JTAG is the default recovery/debug path.

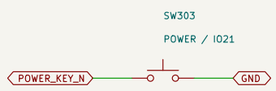


4 USER KEYS / SOFTWARE DEBOUNCE







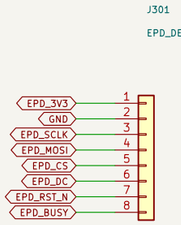


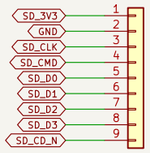


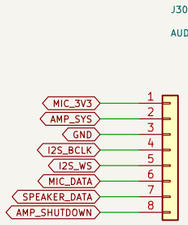


IO21 is the normal sleep/wake key. EN is a separate service control.
I2C pull-ups and IOX_RESET_N pull-down live on POWER_3V3_CONTROL.
EPD / SD / touch / audio signals are global labels to their owning sheets.

2 DEBUG FUNCTION HEADERS / LOAD CIRCUITS PENDING







DEBUG: no external power injection: connect loads only with their domain ON.
Audio: MIC_3V3 and AMP_SYS are different supplies; check module voltage.
Final EPD / SD / audio drivers and their off-state isolation remain pending.
Module pads 1 / 40 / 41 (EPAD) to GND; PCB antenna keepout required.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /MCU_CORE/
File: MCU_CORE.kicad_sch

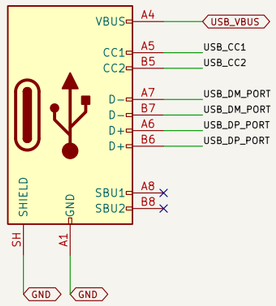
Title: ESP32-S3 N16R8 / RESET / USER KEYS

Size: A3	Date: 2026-11-02	**Rev: r1 DRAFT**
KiCad E.D.A. 10.0.7		Id: 4/7

1 USB-C DEVICE / NO PD / NO HOST POWER

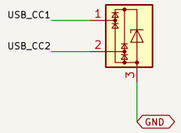
J501

USB-C USB2 16P



U502

TI TPD2E2U06DCKR



USB_CC1

R501

5.1k 1%

GND

USB_CC2

R502

5.1k 1%

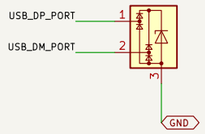
GND

CC1 / CC2 each have their own Rd.
A6+B6 = D+; A7+B7 = D-.
Shield bonded to local GND at connector; final EMC review required.

2 DATA ESD / MCU-SIDE MATCHING

U501

TI TPD2E2U06DCKR



R503

22R





R504

22R





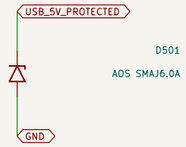
USB_DM_MCU -> GPIO19 / module pad13.
USB_DP_MCU -> GPIO20 / module pad14.
Target 90 ohm differential; short pair, continuous GND return.
USB default 100mA; firmware authorization precedes any 500mA mode.
CC resistors do not negotiate PD, or authorize 1.5A / 3A.
No ESD / USB compliance claim until board-level validation.

3 USB BRANCH PROTECTION / SINGLE-SHUNT SOURCE SELECT

F501

MF-MSMF110/16X-2

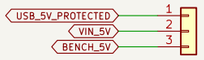






J502

ONE SHUNT: USB / BENCH



1-2 USB / 2-3 BENCH / OPEN = input disconnected.
VIN_5V goes to charger; protected USB is sensed before J502.
One shunt. Known regulated 5V, CURRENT-LIMITED bench input.

PTC / TVS protect the USB branch only: no sustained OVP or automatic source ORing.
TVS leakage / overshoot, fault current, VBUS inrush and cable drop require tests.
Mechanical USB connector MPN / footprint and environmental current ratings are release gates.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /USB_INPUT/
File: USB_INPUT.kicad_sch

Title: USB-C DEVICE / ESD / 5V SOURCE SELECTION

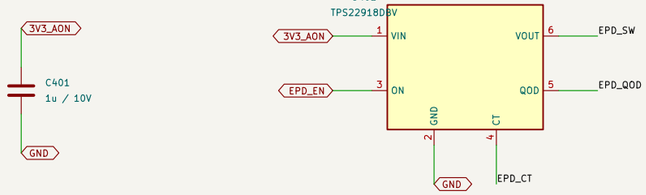
Size: A3	Date: 2026-11-02	**Rev: r1 DRAFT**
KiCad E.D.A. 10.0.7		Id: 5/7

FIVE TPS22918 DOMAINS / COMPONENT VALUES ARE UNVALIDATED CANDIDATES

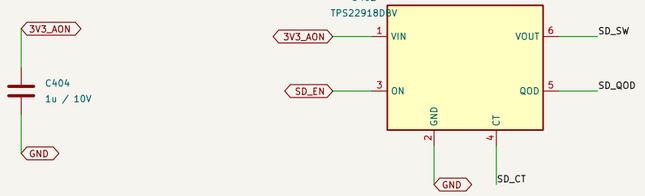
CT = 4.7nF / 25V; RQOD = 330R + internal RPD. OR links measure downstream load only; switch IQ is upstream.

EPD/SD/TOUCH ON pulls: R212/R213/R214 on POWER_3V3_CONTROL. AUDIO_EN: R416 below. No reverse-current blocking.

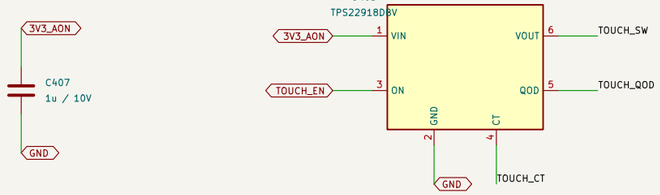
1 EPD



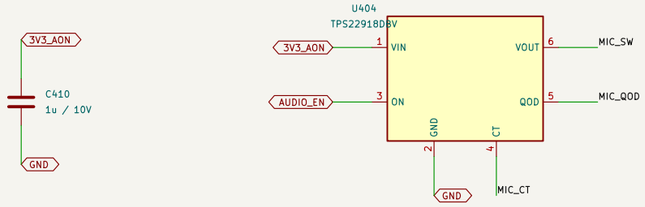
2 SD



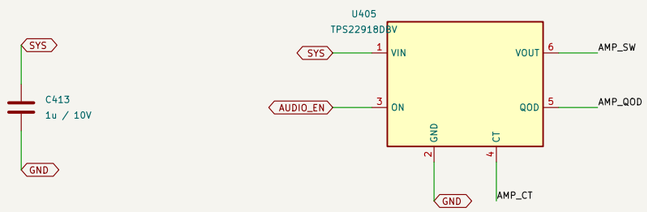
3 TOUCH

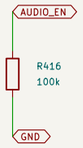


4 MIC



5 AMP





R416: both audio rails default OFF when GPIO4 is high impedance.
SYS may remain alive with AON=0; measure AMP leakage in this state.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_DOMAINS/
File: POWER_DOMAINS.kicad_sch

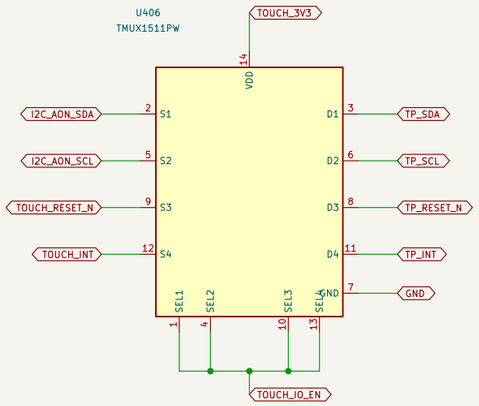
Title: PERIPHERAL POWER / FIVE SWITCHED DOMAINS

Size: A3	Date: 2026-11-02	**Rev: r1 DRAFT**
KiCad E.D.A. 10.0.7		Id: 6/7

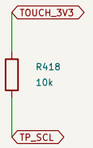
1 TMUX1511PW: ALL FOUR SIGNALS ISOLATED / ACTIVE-HIGH SELECT

VDD, bypass, pull-ups and GT911 VCC all use TOUCH_3V3 AFTER R408 (I_TOUCH). Never power the switch from AON.







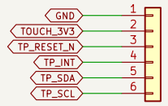




MCU / AON side
GPIO10 SDA; GPIO13 SCL
GPIO14 RESET; GPIO1 INT
R210/R211: 4.7k to AON







R215 = 100k to GND on control sheet.
All SELs LOW: isolated; HIGH: connected.
INT remains bidirectional for GT911 boot.



PWR_FLAG: normal assembly with R408 fitted.
ERC does not model an open measurement link.

6-pin FUNCTIONAL mapping only.
FPC pitch, contact side, latch and footprint await actual sample.

2 POWER / SIGNAL SEQUENCING AND MEASUREMENT

ON: keep TOUCH_IO_EN=0; power TOUCH; wait for valid rail; set MCU RESET/INT; connect TMUX; execute full GT911 reset.
OFF: stop access and touch wake; set TOUCH_IO_EN=0; then TOUCH_EN=0. Do not reset IOX merely on one I2C error.
Start I2C at 100kHz. Pull-up estimate is 3.20k with the present 4.7k and 10k; include actual panel resistors and capacitance.
Powered-off protection: signal pins <=3.6V for VDD<1.5V; verify supply ramps and signal leakage with actual hardware.
With AON=0: R215 must pull SEL LOW even while TOUCH retains charge. GPIO hold/reset behavior is not a power guarantee.

3 LOAD-SWITCH REVIEW / MEASUREMENT LIMITS

Target for each low-voltage load rail: <0.2V within 1s after OFF. EPD high-voltage nodes are outside this target.
CT=4.7nF: TI table ~8ms at 3.3V: formula ~8.6ms (3.3V), ~11.5ms (4.4V). Actual load turn-on and inrush need measurement.
QOD=330R external plus internal RPD. Example only: 200uF, 355R, 3.3V -> 0.2V in ~0.20s with VIN still valid.
Input loss weakens QOD; timing is not guaranteed at VIN=0. Do not backfeed a rail while QOD is discharging it.
Remove measurement OR only AFTER stopping the peripheral and isolating its signals. QOD remains connected on the load side.
TOUCH: opening R408 also removes TMUX VDD; I2C pulls remain on that same isolated load rail. Verify residual voltage first.
External rail injection requires upstream OR AND QOD resistor removal, plus reviewed signal isolation; no direct supply paralleling.

Add actual EPD-driver / SD / audio capacitance before final CT and QOD selection.
SYS -> AMP is a supply gate, not a 5V boost. No load switch provides fault-current limiting.
Refs: TI SLVSD76C pp3,13-15; SCDS390B pp3-4,20-22. Values are NOT bench-validated.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL
ESPaperPlay / xiaobaoaaaaaa

Sheet: /TOUCH_ISOLATION/		
File: TOUCH_ISOLATION.kicad_sch		
Title: GT911 FOUR-SIGNAL ISOLATION / DOMAIN TEST NOTES		
Size: A3	Date: 2026-11-02	**Rev: r1 DRAFT**
KiCad E.D.A. 10.0.7		Id: 7/7